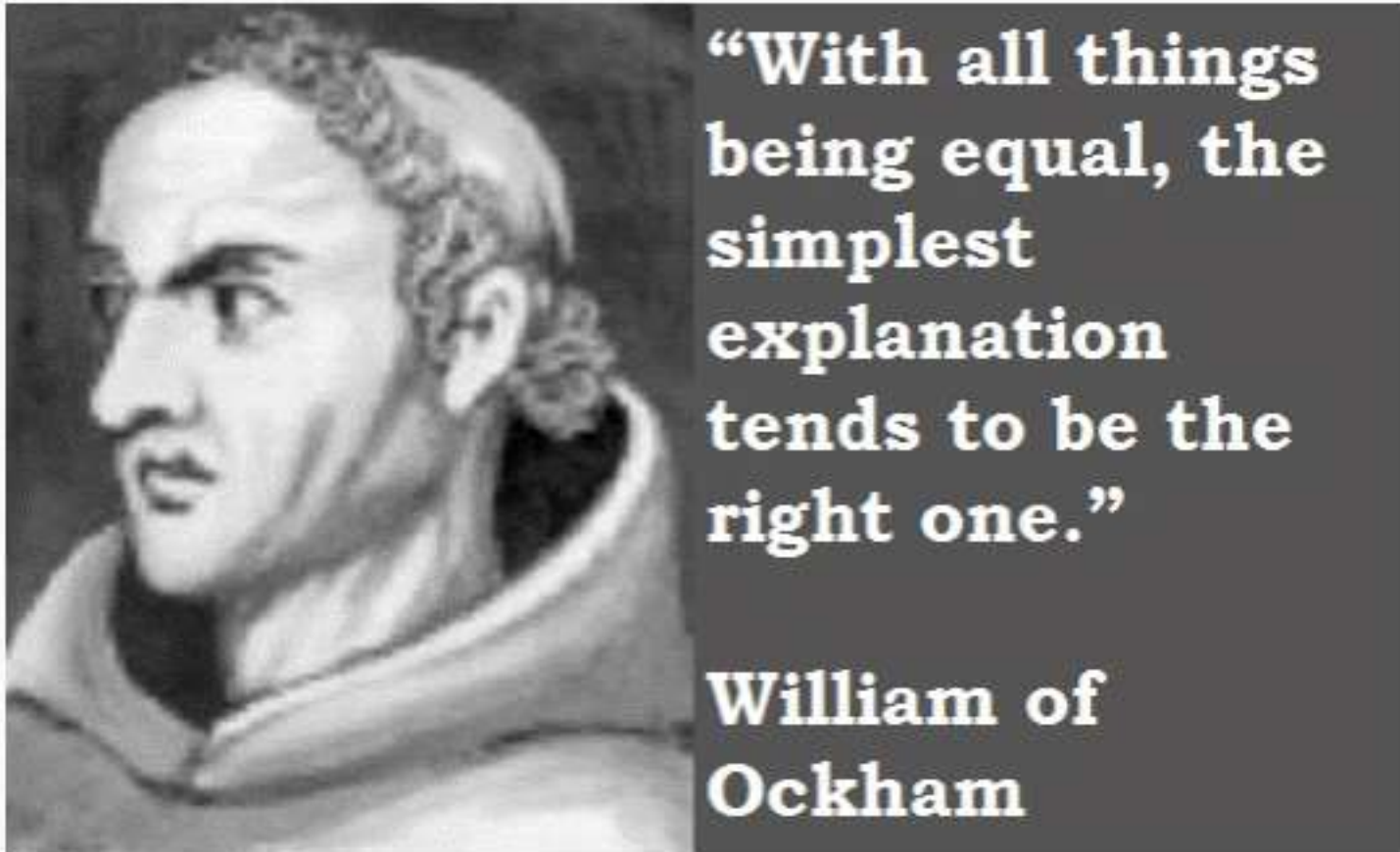


Probabilistic Graphical Models: Undirected Graphs

Statistical Data Mining II
Rachael Hageman Blair

Parsimony Principle/ Occam's razor



Parsimony Principle/ Occam's razor

A rule in science and philosophy stating that entities should not be multiplied needlessly. This rule is interpreted to mean that the simplest of two or more competing theories is preferable and that an explanation for unknown phenomena should first be attempted in terms of what is already known.

Parsimony Principle/ Occam's razor

A rule in science and philosophy stating that entities should not be multiplied needlessly. This rule is interpreted to mean that the simplest of two or more competing theories is preferable and that an explanation for unknown phenomena should first be attempted in terms of what is already known.

*All things being equal,
choose the simplest scientific explanation
that fits the evidence.*

Sparse properties in Graphical Models

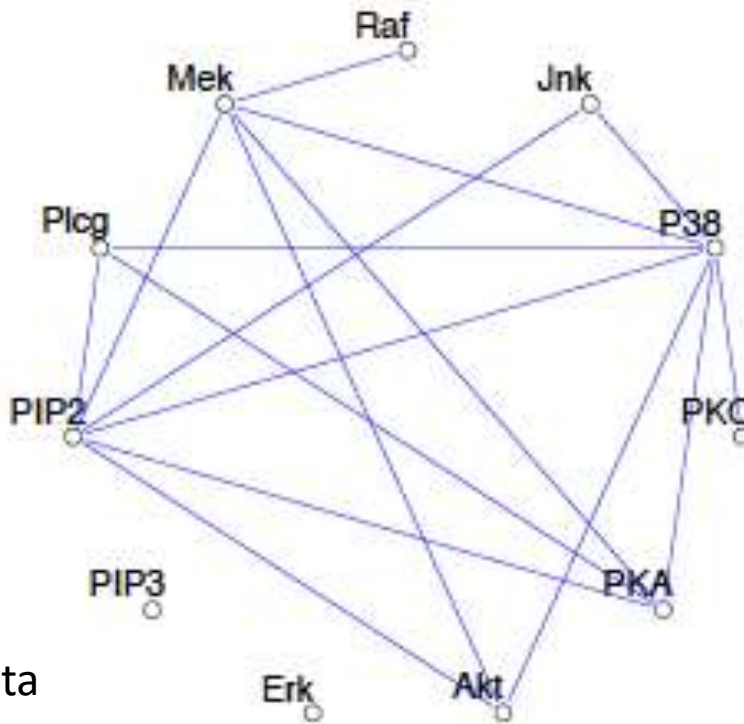
- Sparsity is a highly desirable feature.
- Many attempts to impose sparsity/ feature selection in the inference process.
- Some examples: shrinkage, bootstrap stability selection, informative priors on the structure, etc.

Outline

- Undirected graphs and properties
- Terminology
- Representation
- Relation back to DAGs
- Focus on pairwise graphs
- Focus on continuous data
- Parameter estimation
- Structural Learning
- The graphical lasso

Example: Protein Network

Example of a sparse undirected graph.
Structure estimate via graphical lasso.

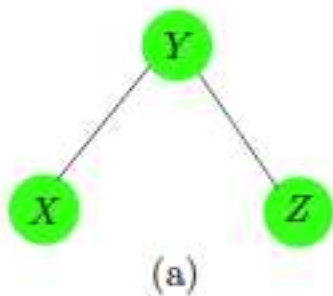


Flow-cytometry Data
N=7466 (cells)
P=11 (proteins)

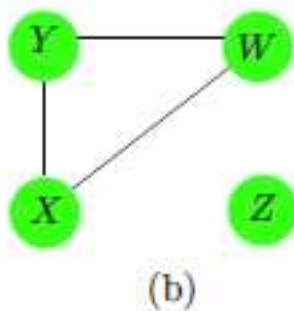
Markov Graphs and Properties

- Same idea as with “directed”, node represent random variable, and edges have a very specific meaning in terms of “conditional independence.

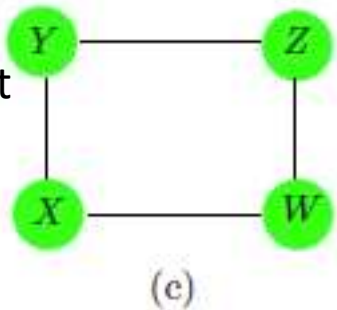
X and Z are Independent given Y.



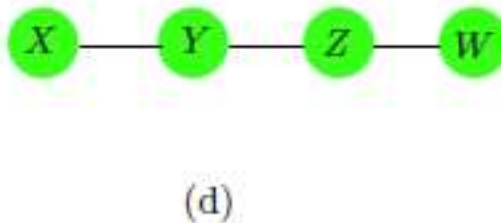
Z is independent Of X, Y, and W.



Z is independent Of X Given Y and W, Etc.



W is independent of Y Given Z.... Etc.



Markov Graphs and Properties

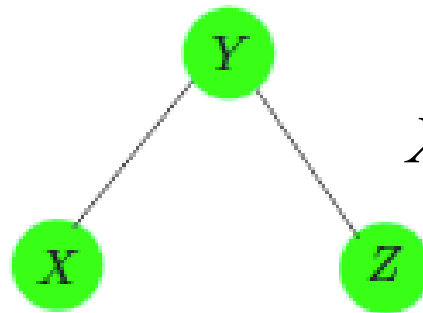
Terminology:

- A graph G consists of a pair (V, E) , where V is a set of vertices and E is the set of edges (defined by pairs of vertices).
- Two vertices X and Y are called **adjacent** if there is an edge joining them, notation: $X \sim Y$.
- A path $X_1, X_2, \dots, X_n$ is a set of vertices that are joined, i.e.,
 $X_{i-1} \sim X_i$ for $i = 1, \dots, n$.
- A complete graph is a graph with every pair of vertices joined by an edge (saturated).

Markov Graphs and Properties

Conditional Independence:

No edge joining X and $Y \longleftrightarrow X \wedge Y \mid (\text{all other vertices})$



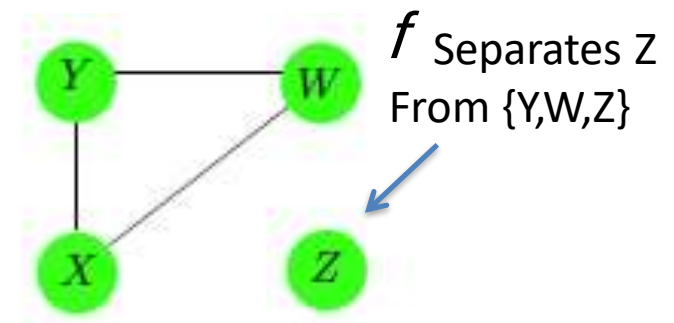
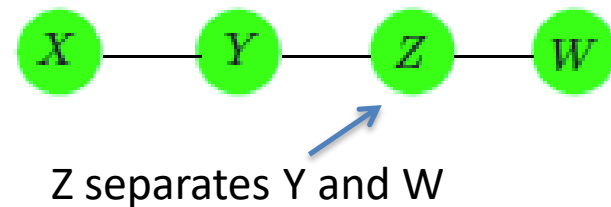
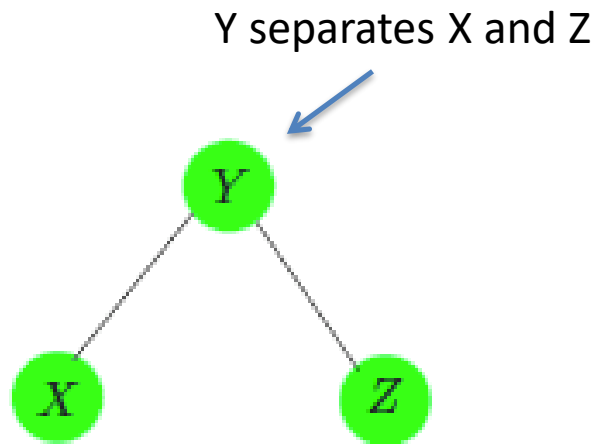
$$X \wedge Z \mid Y$$

A “pairwise” Markov
Independency.

Markov Graphs and Properties

Separation (decompositions):

If A , B , and C are subgraphs, then C is said to separate A and B if every path between A and B intersects a node in C .



Markov Graphs and Properties

Separation (decompositions):

Separation aka decomposition, are useful properties, because they can “break” the graph into conditionally independent pieces.

More formally:

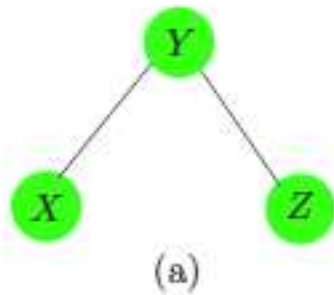
If C separates A and B , then $A \perp B \mid C$.

Markov Graphs and Properties

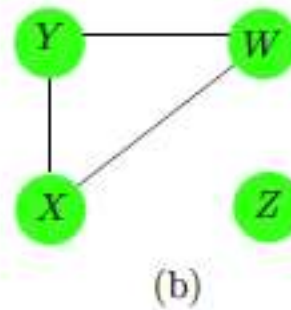
Clique:

A **clique** is a complete subgraph – a set of vertices that are all adjacent to one another, it is called “maximal” if it is a clique and no other vertices can be added to it and still yield a clique.

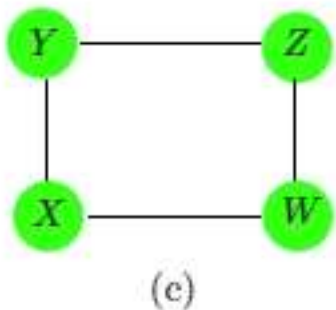
$\{X,Y\},\{Y,Z\}$



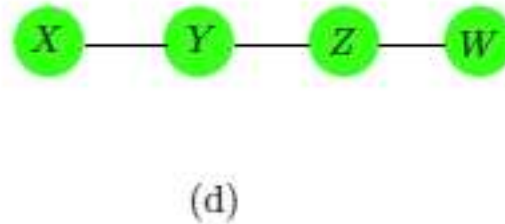
$\{X,Y,W\},\{Z\}$



$\{X,Y\},$
 $\{Y,Z\},$
 $\{Z,W\},$
 $\{X,W\}$



$\{X,Y\}, \{Y,Z\},\{Z,W\}$



Markov Graphs and Properties

Clique:

A **clique** is a complete subgraph – a set of vertices that are all adjacent to one another, it is called “maximal” if it is a clique and no other vertices can be added to it and still yield a clique.

- Many methods for estimation and computation on graphs first decompose the graph into its maximal cliques.
- Relevant quantities are computed in the individual cliques and then accumulated across the entire graph.

Markov Graphs and Properties

Representation:

A probability density function f over a Markov graph G can be represented as:

$$f(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

where $\mathcal{C}$ is the set of maximal cliques, and the positive functions $\psi_C(x_C)$ are the clique potentials.

Recall: these are real valued, not necessarily density functions, which are a special case.

Markov Graphs and Properties

Representation:

A probability density function f over a Markov graph G can be represented as:

$$f(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

where $\mathcal{C}$ is the set of maximal cliques, and the positive functions $\psi_C(x_C)$ are the clique potentials.

Normalizing
Constant, aka
the partition
function

$$Z = \sum_{x \in \mathcal{X}} \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

Markov Graphs and Properties

Representation:

A probability density function f over a Markov graph G can be represented as:

$$f(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi_C(x_C) \quad \text{where} \quad Z = \sum_{x \in \mathcal{X}} \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

where $\mathcal{C}$ is the set of maximal cliques, and the positive functions $\psi_C(x_C)$ are the clique potentials.

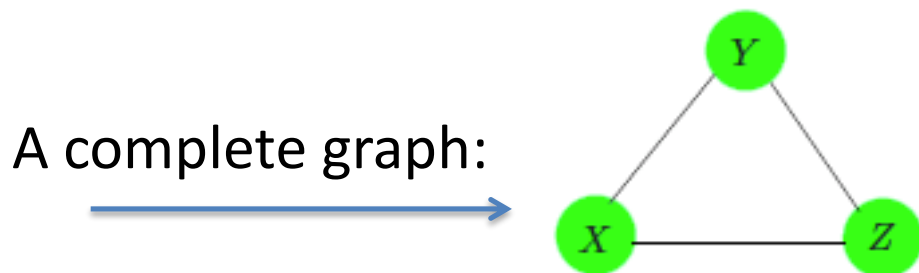
Hammersley-Clifford theorem:

Provides necessary and sufficient conditions in which a positive probability distribution can be represented as a Markov Network (aka a Markov Random Field).

Markov Graphs and Properties

Non-unique representation:

A Markov Network does not always uniquely specify the higher-order dependence structure of a joint probability distribution.



Can have the following distributions:

$$f^{(2)}(x, y, z) = \frac{1}{Z} \psi(x, y) \psi(x, z) \psi(y, z) \quad \text{Pairwise representation}$$

$$f^{(3)}(x, y, z) = \frac{1}{Z} \psi(x, y, z)$$

Markov Graphs and Properties

Note differences between MN and BN:

A Markov Network does not always uniquely specify the higher-order dependence structure of a joint probability distribution.

Recall: A directed acyclic graph, known as a Bayesian Network, can be factorized directly according to the conditional independencies encoded in the graph topology to represent the joint distribution.

Focus: Continuous UDG

Continuous Markov Networks

Assumptions:

- All variables are continuous and Gaussian.
- Specifically, the observations follow a multivariate Gaussian distribution with mean μ and covariance matrix Σ .

Focus: Continuous UDG

Continuous Markov Networks

Assumptions:

- All variables are continuous and Gaussian.
- Specifically, the observations follow a multivariate Gaussian distribution with mean μ and covariance matrix Σ

Our interest: The inverse of the covariance matrix
Is of particular interest.

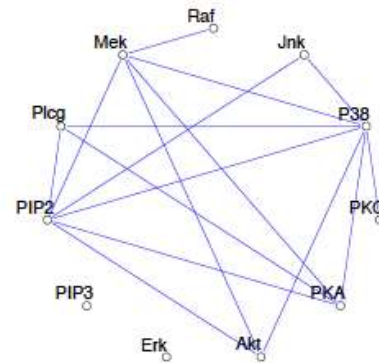
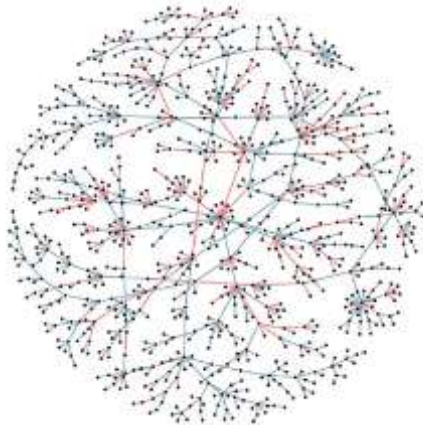
Encodes pairwise information

Focus: Continuous UDG

Continuous Markov Networks

Assumptions:

- All variables are continuous and Gaussian.
- Specifically, the observations follow a multivariate Gaussian distribution with mean μ and covariance matrix Σ .
- The Gaussian distribution encodes at most pairwise information, and it automatically encodes a pairwise Markov graph.



Undirected Graphs

Focus: Continuous UDG

Continuous Markov Networks

Properties:

- All conditional distributions of a Gaussian are also Gaussian.
- Inverse covariance matrix (aka concentration matrix) S^{-1} has a special interpretation about the “partial covariance” between the variables.
- The “partial covariance” between pairs i and j , conditioned on all other variables, is the $(i, j)^{\text{th}}$ element of $Q = S^{-1}$.
- If the $(i, j)^{\text{th}}$ element of $Q = S^{-1}$ is zero, then variables i and j are conditionally independent, given the other variables. ★

Focus: Continuous UDG

- This formalism has a name: the **Gaussian Graphical Model** (GGM).
- The estimation problem is a “natural one”, which concerns the estimation of the inverse of the covariance matrix, to extract conditional independencies/ and dependencies in a continuous pairwise Markov Network $Q = S^{-1}$.

BIOMETRICS 28, 157-75
March 1972

COVARIANCE SELECTION

A. P. DEMPSTER

Department of Statistics, Harvard University, Cambridge, Massachusetts 02138, U. S. A.

SUMMARY

The covariance structure of a multivariate normal population can be simplified by setting elements of the inverse of the covariance matrix to zero. Reasons for adopting such a model and a rule for estimating its parameters are given in section 2. It is also proposed to select the zeros in the inverse from sample data. A numerical illustration of the proposed technique is given in section 3. Appendix A sketches the general theory of exponential families which underlies the special results of section 2, and Appendix B describes two approaches to computation of the proposed estimator.

Focus: Continuous UDG

An illustrative example:

Suppose we have $\{X_1, X_2, \dots, X_{p-1}, X_p\}$

That is we partition: $X = (Y, Z)$.

The covariance can be block partitioned accordingly:

$$S = \begin{pmatrix} S_{ZZ} & S_{ZY} \\ S_{ZY}^T & S_{YY} \end{pmatrix}$$

Focus: Continuous UDG

An illustrative example (con't):

The conditional distribution of Y given Z is given as:

$$Y | Z \sim N\left(\underbrace{m_Y + (z - m_Z)^T S_{ZZ}^{-1} s_{ZY}}_{\text{Conditional Mean}}, \underbrace{s_{YY} - s_{ZY}^T S_{ZZ}^{-1} s_{ZY}}_{\text{Variance}}\right)$$

Conditional Mean has exactly the
Multivariate linear regression of Y on Z .

Regression coefficient

Focus: Continuous UDG

Lets think about the inverse of the covariance matrix:

If we partition $Q = S^{-1}$ in the same way, it can be shown using formulas for partitioned inverses, that:

$$Q_{ZY} = -Q_{YY} \times \underline{S_{ZZ}^{-1} s_{ZY}},$$

where $1/Q_{YY} = s_{YY} - s_{ZY}^T S_{ZZ}^{-1} s_{ZY} > 0$.

Therefore, we can write:

$$\begin{aligned} b &= S_{ZZ}^{-1} s_{ZY} \\ &= -q_{ZY} / q_{YY}. \end{aligned}$$

Contains the dependence Information.

Parameter Estimation

Problem formulation

Given some realizations (data), X we want to estimate the parameters of a UDG that approximates the joint distribution.

Suppose we assume the graph is complete (fully connected), we can write the sample covariance as:

$$S = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})^T .$$


Sample mean

Ignoring constants, we can write down the log-likelihood.

Parameter Estimation

Log likelihood:

$$l(Q) = \log \det(Q) - \text{trace}(SQ).$$

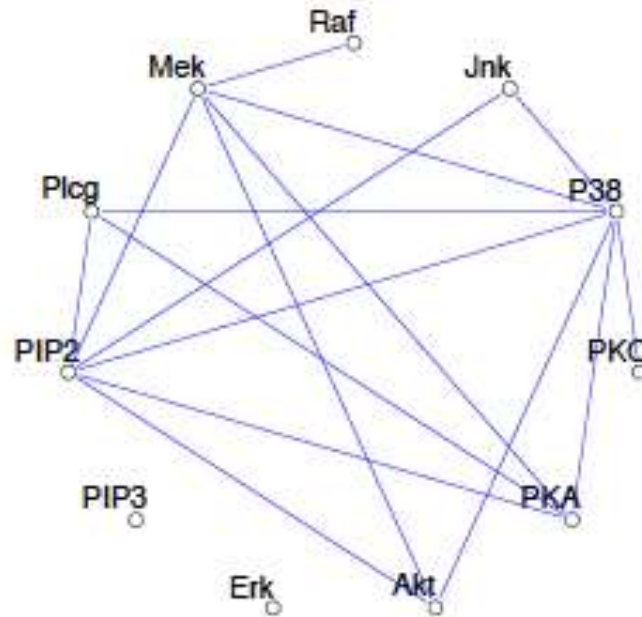
- The quantity $-l(Q)$ is a convex function of Q .

Complete graphs, difficult to interpret – especially in high dimensions. Let's think about constraints.

Parameter Estimation

Sparsity Constraints

Lets assume that some of the edges are missing.



$$S^{-1}(Raf, Jnk) = 0$$

$$S^{-1}(Raf, P38) = 0$$

⋮

Parameter Estimation

Sparsity Constraints

The new optimization is “constrained”.

In other words, solve:

$$\max l(Q) = \log \det(Q) - \text{trace}(SQ).$$

Subject to the constraint that some pre-defined set of entries in

$Q = S^{-1}$ are equal to zero.

Parameter Estimation

Sparsity Constraints

The new optimization is “constrained”.

In other words, solve:

$$\max l(Q) = \log \det(Q) - \text{trace}(SQ).$$

Subject to the constraint that some pre-defined set of entries in

$Q = S^{-1}$ are equal to zero.

“Equality constraint optimization” is also convex, and has been solved e.g., iterative proportional fitting procedure (Speed and Kiiveri, 1986), and other methods summarize in (Lauritzen, 1996).

Parameter Estimation

Alternative Approach to Constrained Optimization:

- Rooted in regression ideas.
- Suppose we want to estimate edge parameters $q_{ij} \hat{=} Q$, for those vertices that are joined to a given vertex i , and restricting those that are not joined to be 0.

Intuition: Regression of i on other variables may provide a reasonable estimate.

Problem: Ignores the dependence structure among the other variables.

Parameter Estimation

Alternative Approach to Constrained Optimization:

Modify the equation:

$$l(Q) = \log \det(Q) - \text{trace}(SQ) - \sum_{(j,k) \notin E} g_{jk} q_{jk}.$$

Lagrange constants for missing edges

The derivative can be given as:

$$\frac{dL}{dQ} = Q^{-1} - S - G$$

Matrix of Lagrange parameters.

We can use a regression to find the solution!

Parameter Estimation

Alternative Approach to Constrained Optimization:

$$Q^{-1} - S - G = 0$$

We can use regression to solve for Q and $W = Q^{-1}$ one row and one column at a time.

Using the same rationale as before, we can use a “block partition”. The upper right block, for example,

$$w_{12} - s_{12} - g_{12} = 0.$$

Parameter Estimation

Alternative Approach to Constrained Optimization:

Now, subbing into: $w_{12} - s_{12} - g_{12} = 0$,

we have $w_{12}b - s_{12} - g_{12} = 0$,

Which is a linear system, with $p-1$ equations for the constrained regression of X_p on the other predictors.

We can reduce this a little bit... note a nonzero element in g_{12} means that an edge is constrained to be zero. We can reduce the system accordingly.

Parameter Estimation

Alternative Approach to Constrained Optimization:

We can also partition Q and $W = Q^{-1}$:

$$\begin{bmatrix} W_{11} & w_{12} \\ w_{12}^T & w_{22} \end{bmatrix} \begin{bmatrix} Q_{11} & q_{12} \\ q_{12}^T & q_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0^T & 1 \end{bmatrix}$$

This implies:

$$\begin{aligned} w_{12} &= -W_{11}q_{12} / q_{22} \quad \star \\ &= W_{11}b \end{aligned}$$

(similar to as before), where $b = -q_{12} / q_{22}$

Parameter Estimation

Alternative Approach to Constrained Optimization:

If we remove those elements, and subset the rest of the equation accordingly (take off the zero elements piece), we have the following reduced set of equations:

$$W_{11}^* b^* - s_{12}^* = 0,$$

with solution: $\hat{b}^* = W_{11}^{*-1} s_{12}^*$, which is clearly true up to a constant (Recall  $w_{12} = -W_{11} q_{12} / q_{22}$). But the constant can be shown to

$$= W_{11} b$$

Be $1/q_{22} = w_{22} - w_{12}^T b$, and $w_{22} = s_{22}$.

Parameter Estimation

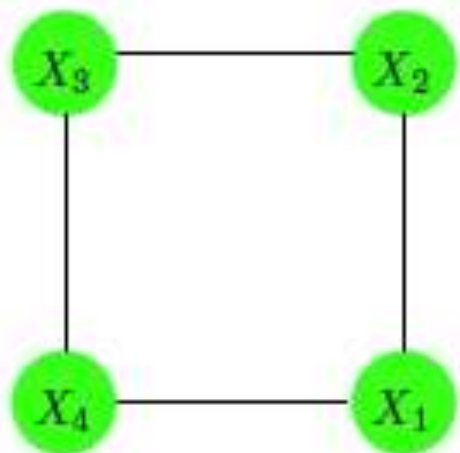
- Note that the regressions are coupled (not independent).
- This coupling is through the use of a common W .

Parameter Estimation

Algorithm 17.1 *A Modified Regression Algorithm for Estimation of an Undirected Gaussian Graphical Model with Known Structure.*

1. Initialize $\mathbf{W} = \mathbf{S}$.
 2. Repeat for $j = 1, 2, \dots, p, 1, 2, \dots, p, \dots$ until convergence:
 - (a) Partition the matrix $\mathbf{W}$ into part 1: all but the j th row and column, and part 2: the j th row and column.
 - (b) Solve $\mathbf{W}_{11}^* \beta^* - s_{12}^* = 0$ for the unconstrained edge parameters β^* , using the reduced system of equations as in (17.19). Obtain $\hat{\beta}$ by padding $\hat{\beta}^*$ with zeros in the appropriate positions.
 - (c) Update $w_{12} = \mathbf{W}_{11} \hat{\beta}$
 3. In the final cycle (for each j) solve for $\hat{\theta}_{12} = -\hat{\beta} \cdot \hat{\theta}_{22}$, with $1/\hat{\theta}_{22} = s_{22} - w_{12}^T \hat{\beta}$.
-

Parameter Estimation



$$\mathbf{S} = \begin{pmatrix} 10 & 1 & 5 & 4 \\ 1 & 10 & 2 & 6 \\ 5 & 2 & 10 & 3 \\ 4 & 6 & 3 & 10 \end{pmatrix}$$

FIGURE 17.4. A simple graph for illustration, along with the empirical covariance matrix.

Solution converges quickly:

$$\hat{\Sigma} = \begin{pmatrix} 10.00 & 1.00 & 1.31 & 4.00 \\ 1.00 & 10.00 & 2.00 & 0.87 \\ 1.31 & 2.00 & 10.00 & 3.00 \\ 4.00 & 0.87 & 3.00 & 10.00 \end{pmatrix}, \quad \hat{\Sigma}^{-1} = \begin{pmatrix} 0.12 & -0.01 & 0.00 & -0.05 \\ -0.01 & 0.11 & -0.02 & 0.00 \\ 0.00 & -0.02 & 0.11 & -0.03 \\ -0.05 & 0.00 & -0.03 & 0.13 \end{pmatrix}$$

Note the zeroes in $\hat{\Sigma}^{-1}$, corresponding to the missing edges (1,3) and (2,4).

Graph Structure Estimation

- The more common problem: to learn the structure from the data.
- One approach (Meinshausen and Bühlmann, 2006): instead of estimating S or $Q = S^{-1}$ fully and directly, the instead only estimate which components of q_{ij} are nonzero.
 - A lasso regression is fit to each variable, where $Y = q_i$, is played by each variable, and the $X \hat{\Gamma} \{q_{j^1 i}\}$.
 - An edge exists $q_{ij} \neq 0$, if the regression of X_i on $Y = X_j$ is nonzero, or the regression of X_j on $Y = X_i$.

Note: This is NOT the graphical LASSO.

Shrinkage Methods: The Lasso

- The lasso estimate is defined as:

$$\hat{b}^{lasso} = \underset{b}{\operatorname{argmin}} \sum_{i=1}^N (y_i - b_0 - \sum_{j=1}^p x_{ij} b_j)^2$$

subject to:

$$\sum_{j=1}^p |b_j| \leq t.$$

L1 lasso penalty

- Equivalently:

$$\hat{b}^{lasso} = \underset{b}{\operatorname{argmin}} \left[\frac{1}{2} \sum_{i=1}^N (y_i - b_0 - \sum_{j=1}^p x_{ij} b_j)^2 + \lambda \sum_{j=1}^p |b_j| \right]$$

- Quadratic programming problem.
- Shares properties of ridge regression and subset selection.



Graph Structure Estimation

Graphical LASSO, is a different approach, which puts a penalty on the (proportional) likelihood:

$$\log \det Q - \frac{1}{2} \text{trace}(SQ) - \frac{\lambda}{2} \|Q\|_1,$$

Where $\|Q\|_1$ is the L1 norm – the sum of absolute values of the elements of $Q = S^{-1}$.

- The optimization is convex.
- Can work within the earlier algorithm (17.1) for the iterative procedure, many have been proposed (e.g., pathwise coordinate descent).
- Efficient... sparse inference with 1000 nodes, less than one minute.

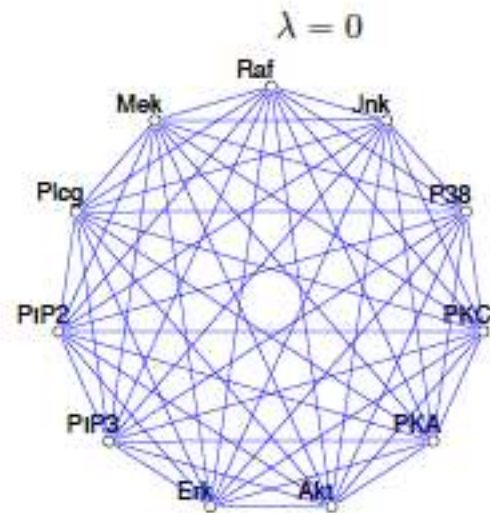
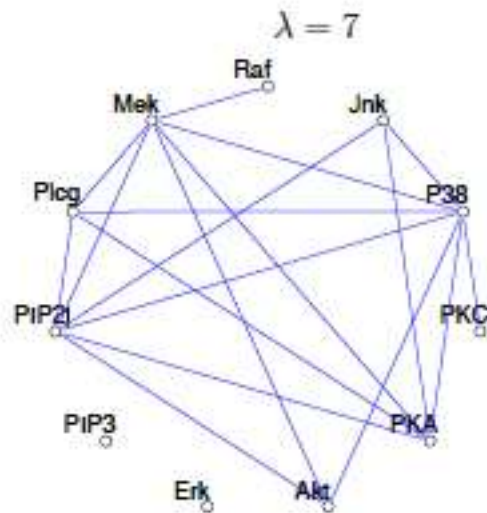
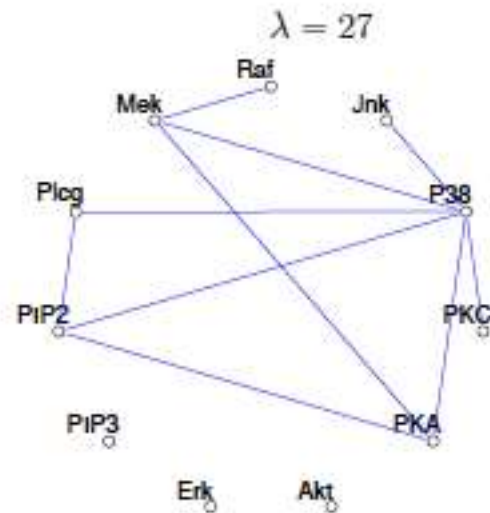
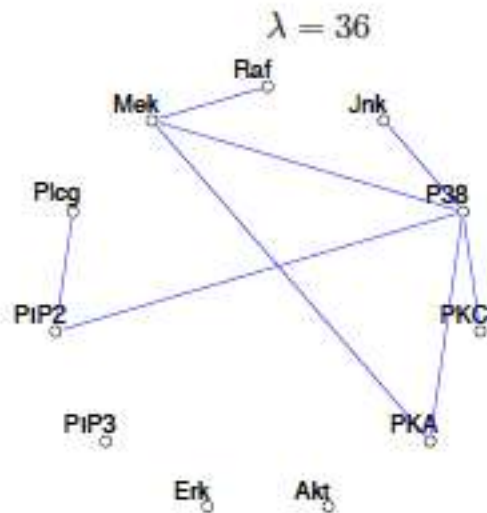
Note: it is not a pairwise!

Graph Structure Estimation

Algorithm 17.2 *Graphical Lasso.*

1. Initialize $\mathbf{W} = \mathbf{S} + \lambda \mathbf{I}$. The diagonal of $\mathbf{W}$ remains unchanged in what follows.
 2. Repeat for $j = 1, 2, \dots, p, 1, 2, \dots, p, \dots$ until convergence:
 - (a) Partition the matrix $\mathbf{W}$ into part 1: all but the j th row and column, and part 2: the j th row and column.
 - (b) Solve the estimating equations $\mathbf{W}_{11}\beta - s_{12} + \lambda \cdot \text{Sign}(\beta) = 0$ using the cyclical coordinate-descent algorithm (17.26) for the modified lasso.
 - (c) Update $w_{12} = \mathbf{W}_{11}\hat{\beta}$
 3. In the final cycle (for each j) solve for $\hat{\theta}_{12} = -\hat{\beta} \cdot \hat{\theta}_{22}$, with $1/\hat{\theta}_{22} = w_{22} - w_{12}^T \hat{\beta}$.
-

Graph Structure Estimation



Graph Structure Estimation

Other points on Graphical Lasso:

- Can also trace the solution paths, as a function of the penalty parameter λ .
- Can compute over a range of lambdas.
- Model selection difficult, but can guide in feature selection.

Sparse inverse covariance estimation with the graphical lasso

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Graphical lasso in R and Matlab:

Glasso: Graphical lasso for R

Authors: Jerome Friedman, Trevor Hastie and Rob Tibshirani
Maintainer: Rob Tibshirani

This software, written in the R language, estimates a sparse inverse covariance matrix using a lasso (L1) penalty. It can be used for estimating a sparse undirected graph. Version 1.3 fixed some errors in the Meinhausen-Buhlmann approximation, and includes a new function for computing the solution along a path of regularization parameters.

Based on the paper
Jerome Friedman, Trevor Hastie and Robert Tibshirani [Sparse inverse covariance estimation with the graphical lasso](#), *Biostatistics*, December 12, 2007.

[Linux, Windows and MAC versions available at CRAN](#)

Matlab package

[Zip file](#) Maintained by Hossein Karshenas: hosseinkarshenas@gmail.com

Conclusions

- Gaussian graphical models are defined by the inverse of the covariance matrix of the data.
- These are pairwise Markov Networks for continuous data.
- Two problems: parameter and structural learning. 99% of the time, it's the structure that we are after.
- Sparsity can facilitate model interpretation.
- Model selection is difficult.
- Ideally, one visualizes the structures over a range of complexity parameters and summarizes features accordingly.